

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. Examination, 2024
MATH – 1105, Differential and Integral Calculus

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) Sketch the function $f(x) = \sqrt{x-1}$ 3
 b) If the function $f(x) = \begin{cases} x^2 + 1 & \text{when } x \leq 0 \\ 1 + \sin x & \text{when } 0 < x < 1 \\ 2 + (x - \frac{\pi}{2})^2 & \text{when } 1 \leq x \end{cases}$ 5
 then discuss the differentiability at $x = 1$.
 c) Find the tangent line and normal line of $y = x^2 + 5x$ at $x = -1$ 6
2. a) Solve the following by using L' hospitals rule: 6
 (i) $\lim_{x \rightarrow 0} [\frac{1}{x^2} - \frac{1}{\sin^2 x}]$ (ii) $\lim_{x \rightarrow 1} \frac{1 + \cos \pi x}{x^2 - 2x + 1}$ (iii) $\lim_{x \rightarrow \infty} x^2 e^{-x}$
 b) Find the differential co-efficient of the following: 6
 (i) $\tan^{-1}(\frac{x}{\sqrt{1-x^2}})$ (ii) $\sin^{-1}(\frac{2x}{1+x^2})$ (iii) $x^{\log x}$
 c) Find out the Maclaurin's series of $1/(1-x)$ 2,
3. a) Define Gamma function. 2
 b) Prove that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ 6
 c) Find the perimeter of the circle $x^2 + y^2 = a^2$ 6
4. a) State the Mean Value theorem and If $f(a+h) - f(a) = hf'(a + \theta h)$; $0 < \theta < 1$, 4
 when $h = 3, a = 0$; $f(x) = \frac{1}{3}x^3 - \frac{3}{2}x^2 + 2x$; Show that θ has got two values find them.
 b) Find the relative maxima and minima of $f(x, y) = x^3 + y^3 - 3x - 12y + 20$ 5
 c) Find the area of the larger portion enclosed by the curves $y^2 = 2ax - x^2$ and $y^2 = ax$. 5
5. a) State and prove Leibnitz's theorem and verify Mean Value theorem for $f(x) = 3 + 2x - x^2$ 7
 over $(0, 1)$
 b) A rectangular box, open at the top, is to have a volume of 32 cubic feet. What must be the dimensions so that the total surface is a minimum? 3
 c) Find n-th derivatives of the following function: (i) $y = \frac{1}{(x+a)}$ (ii) $y = \ln(ax + b)$ 4
6. a) Find the volume of the solid formed by the revolution of the ellipse 5
 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ about x-axis.
 b) Estimate the maximum value, minimum value, critical point of the function, 4
 $x^3 - 9x^2 + 15x - 3$
 c) A point moves on the parabola $3y = x^2$ in such a way that when $x = 3$, the abscissa is 3
 increasing at the rate of 3cm per second. At what rate is the ordinate increasing at the point?
 d) Find $(f \circ g \circ h)(x)$ if $f(x) = \sqrt{x}$, $g(x) = \frac{1}{x}$ and $h(x) = x^3$ 2
7. a) Evaluate the given integrals: 9
 i) $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\tan x}}{1 + \sqrt{\tan x}} dx$ ii) $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$ iii) $\int_0^1 x^3 \sqrt{1 + 3x^4} dx$
 b) Define Beta Function. Evaluate: i) $\int \sin^4 x \cos^3 x dx$ ii) $\int \frac{dx}{\sqrt{(x-a)(b-x)}}$ 5